

Develop Scientific Skills

Scientific Skills Exercise

Analyzing Polypeptide Sequence Data

Are Rhesus Monkeys or Gibbons More Closely Related to Humans? In this exercise, you will look at amino acid sequence data for the β -polypeptide chain of hemoglobin, often called β -globin. You will then interpret the data to hypothesize whether the monkey or the gibbon is more closely related to humans.

How Such Experiments Are Done Researchers can isolate the polypeptide of interest from an organism and then determine the amino acid sequence. More frequently, the DNA of the relevant gene is sequenced, and the amino acid sequence of the polypeptide is deduced from the DNA sequence of its gene.

Data from the Experiments In the data below, the letters give the sequence of the 146 amino acids in β -globin from humans, rhesus monkeys, and gibbons. Because a complete sequence would not fit on one line here, the sequences are divided into three segments: amino acids 1–50, 51–100, and 101–146. The sequences for the three different species are aligned so that you can compare them easily. For example, you can see that for all three species, the first amino acid is V (valine) and the 146th amino acid is H (histidine).

INTERPRET THE DATA

- Scan the monkey and gibbon sequences, letter by letter, finding any amino acids that do not match the human sequence. (a) How many amino acids differ between the monkey and the human sequences? (b) Between the gibbon and human?
- For each nonhuman species, what percent of its amino acids are identical to the human sequence of β -globin?
- Based on these data alone, state a hypothesis for which of these two species is more closely related to humans. What is your reasoning?
- What other evidence could you use to support your hypothesis?

Instructors: A version of this Scientific Skills Exercise can be assigned in Mastering Biology.

Species	Alignment of Amino Acid Sequences of β -globin		
Human	1	51	101
Monkey	VHLTPEEKSAVTALMGKLVVDEVGAEALGRLLVVYPHTQRFFESFGLST	VLTALMGKLVVDEVGAEALGRLLVVYPHTQRFFESFGLST	ENRLLGGVIVICVLAHHFGREFTPPVGNAYQKVMGVAVALAHKTH
Gibbon	VHLTPEEKSAVTALMGKLVVDEVGAEALGRLLVVYPHTQRFFESFGLST	VLTALMGKLVVDEVGAEALGRLLVVYPHTQRFFESFGLST	ENRLLGGVIVICVLAHHFGREFTPPVGNAYQKVMGVAVALAHKTH

Data from Hargill, <http://www.ncbi.nlm.nih.gov/pubmed/1111111>; rhesus monkey, Miki, <http://www.ncbi.nlm.nih.gov/pubmed/1222222>; gibbon, <http://www.ncbi.nlm.nih.gov/pubmed/1333333>.

Scientific Skills Exercises in every chapter of the text use real data to build key skills needed for biology, including data analysis, graphing, experimental design, and math skills. Each exercise is also available as an automatically graded assignment in Mastering Biology with answer-specific feedback for students.

Problem-Solving Exercises guide students in applying scientific skills and interpreting real data in the context of solving a real-world problem. A version of each Problem-Solving Exercise can also be assigned in Mastering Biology.

PROBLEM-SOLVING EXERCISE

Can declining amphibian populations be saved by a vaccine?

Amphibian populations are declining rapidly worldwide. The fungus *Batrachochytrium dendrobatidis* (*Bd*) has contributed to this decline. This pathogen causes severe skin infections in many amphibian species, leading to massive die-offs. Efforts to save amphibians from *Bd* have had limited success, and there is little evidence that frogs and other amphibians have acquired resistance to *Bd* on their own.



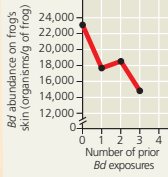
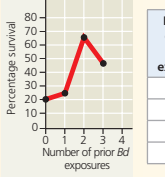
Yellow-legged frogs (*Rana muscosa*) in California killed by *Bd* infection

Instructors: A version of this Problem-Solving Exercise can be assigned in Mastering Biology.

In this exercise, you will investigate whether amphibians can acquire resistance to the fungal pathogen *Bd*.

Your Approach The principle guiding your investigation is that prior exposure to a pathogen can enable amphibians to acquire immunological resistance to that pathogen. To see whether this occurs after exposure to *Bd*, you will analyze data on acquired resistance in Cuban tree frogs (*Osteopilus septentrionalis*).

Your Data To create variation in number of prior exposures to *Bd*, Cuban tree frogs were exposed to *Bd* and cleared of their infection (using heat treatments) from zero to three times; frogs with no prior exposures are referred to as "naïve." Researchers then exposed frogs to *Bd* and measured mean abundance of *Bd* on the frog's skin, frog survival, and abundance of lymphocytes (a type of white blood cell involved in the vertebrate immune response).

Number of prior <i>Bd</i> exposures	Thousands of lymphocytes per g of frog
0	134
1	240
2	244
3	227

Your Analysis

- Describe and interpret the results shown in the figure.
- (a) Graph the data in the table. (b) Based on these data, develop a hypothesis that explains the results discussed in question 1.
- Breeding populations of amphibian species threatened by *Bd* have been established in captivity. In addition, evidence suggests that Cuban tree frogs can acquire resistance after exposure to dead *Bd*. Based on this information and your answers to questions 1 and 2, suggest a strategy for repopulating regions decimated by *Bd*.